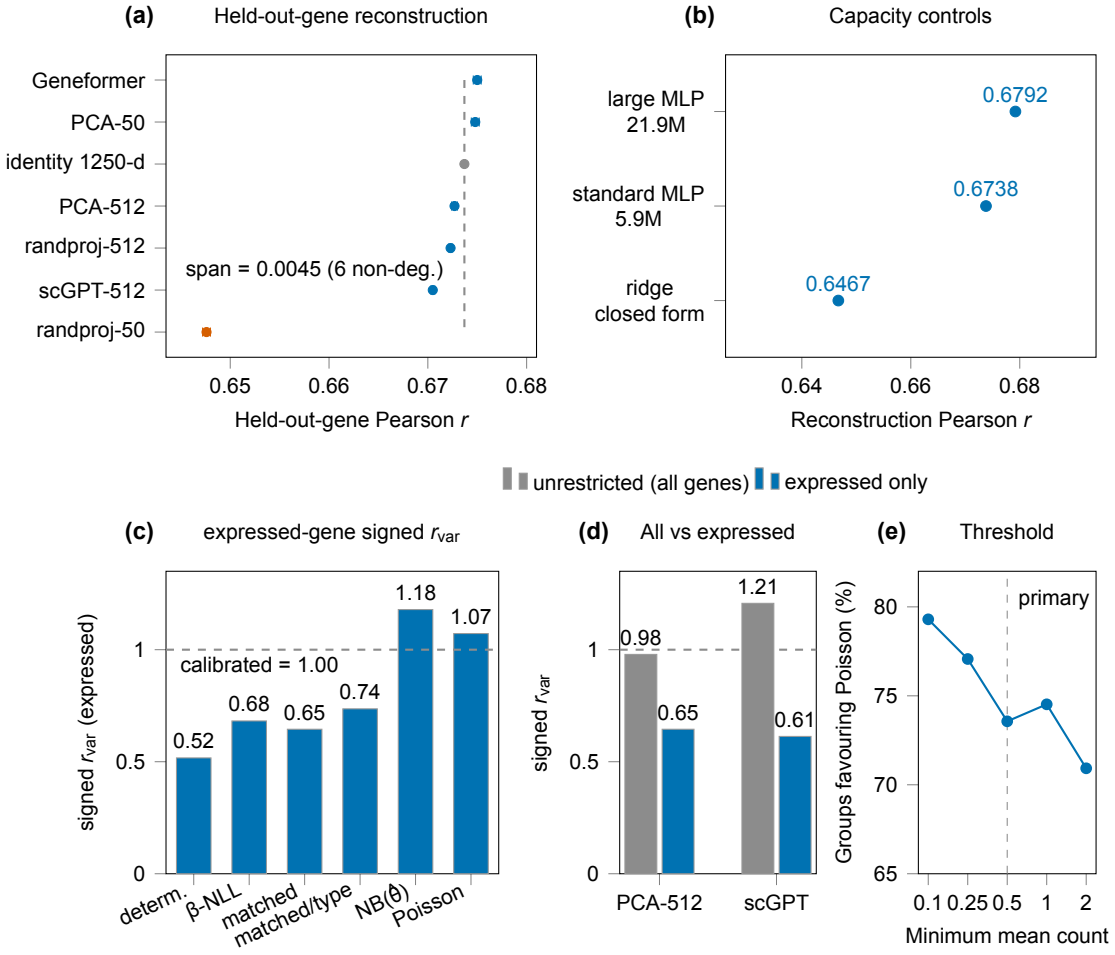
(a)
Held-out-gene reconstruction
Geneformer
PCA-50
identity 1250-d
PCA-512
randproj-512
scGPT-512
randproj-50
span = 0.0045 (6 non-deg.)
0.65
0.66
0.67
0.68
Held-out-gene Pearson r
(b)
Capacity controls
large MLP
21.9M
standard MLP
5.9M
ridge
closed form
0.6792
0.6738
0.6467
0.64
0.66
0.68
Reconstruction Pearson r
unrestricted (all genes)
expressed only
(c)
expressed-gene signed r_var
signed r_var (expressed)
calibrated = 1.00
0.52
0.68
0.65
0.74
1.18
1.07
determ.
β-NLL
matched
matched/type
NB(θ)
Poisson
(d)
All vs expressed
signed r_var
0.98
0.65
1.21
0.61
PCA-512
scGPT
(e)
Threshold
Groups favouring Poisson (%)
primary
80
75
70
65
0.1
0.25
0.5
1
2
Minimum mean count